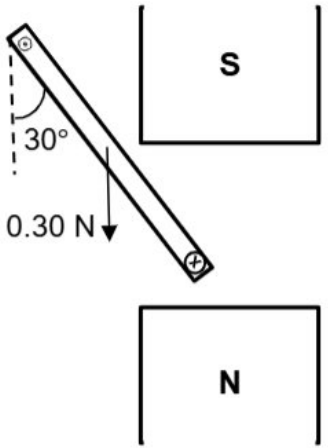
4 A rigid square coil of sides 50 mm has 100 turns. It hangs vertically such that it can rotate freely about one side.

When a current of 0.40 A passes through the coil, it hangs in equilibrium at an angle of 30° to the vertical with its lowest side in the vertical magnetic field of a magnet. The weight of the coil is 0.30 N.



What is the flux density of the field between the magnets?

A 0.038 T **B** 0.043 T **C** 0.075 T **D** 0.087 T